Colorectal-Specific Radiation Dose and Chemotherapy Risk for Subsequent Colorectal Malignancies in Childhood Cancer Survivors: A Childhood Cancer Survivor Study (CCSS) Report

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ABSTRACT

PURPOSE Among childhood cancer survivors, we evaluated not previously explored relationships between colorectal subsequent malignant neoplasm (SMN) incidence and colorectum-specific radiation dose metrics currently used in radiation therapy (RT) planning and expanded upon previously reported chemotherapy associations.

METHODS The Childhood Cancer Survivor Study (CCSS) includes 5-year survivors of childhood cancer diagnosed between 1970 and 1999. RT was assessed as mean colorectal dose (MCD) and the percent volume ($V_{X\ Gy}$) receiving ≥5, 10, 20, 30, and 40 Gy. Chemotherapy was assessed as cumulative doses for procarbazine and platinum agents, cyclophosphamide-equivalent doses for alkylating agents, and doxorubicin-equivalent doses for anthracyclines. Piecewise-exponential models and excess rate ratio (ERR) models evaluated dose-response relationships for the incidence of colorectal SMNs. Reference groups were those not receiving the assessed treatment(s).

RESULTS Among 25,723 survivors (median follow-up = 28.5 years; range = 5.0-48.9), 104 colorectal SMNs were identified. A dose-response relationship was observed between MCD and colorectal SMN rates; incidence rate ratios (IRRs) for 10 to <20 Gy and ≥20 Gy were 3.6 (95% CI, 1.9 to 6.9) and 8.3 (95% CI, 3.9 to 17.8), respectively. When ≥20% of the colorectum volume was irradiated, IRRs increased with increasing volume. The $V_{20\ Gy}$ IRRs were 3.8 (95% CI, 1.9 to 7.6), 4.9 (95% CI, 2.0 to 12.0), and 8.7 (95% CI, 3.5 to 21.6) for irradiated volumes of 20% to <40%, 40% to <80%, and ≥80%, respectively. The IRR was 1.8 (95% CI, 1.0 to 3.0) for doxorubicin-equivalent dose ≥250 mg/m², 3.7 (95% CI, 2.2 to 6.4) for cyclophosphamide-equivalent dose ≥6,000 mg/m², and 4.5 (95% CI, 2.0 to 10.1) for platinum dose ≥450 mg/m². For procarbazine dose, the IRR was 6.3 (95% CI, 3.0 to 13.2) for 4,200 to <7,036 mg/m² and 9.0 (95% CI, 4.3 to 18.9) for ≥7,036 mg/m². In the absence of RT, colorectal SMN rates increased with exposure to any platinum-based agent (IRR, 3.8 [95% CI, 1.1 to 12.7]), alkylator (IRR, 4.8 [95% CI, 1.6 to 14.4]), or procarbazine (IRR, 16.9 [95% CI, 5.9 to 48.8]). Colorectal SMN rates increased linearly with procarbazine dose (ERR per 1,000 mg/m² = 73.0 [95% CI, 26.4% to 119.6%]) and MCD (ERR per 1 Gy = 20.8 [95% CI, 9.0% to 32.5%]). Quadratic ERR models did not improve data fit compared with linear ERR models.

CONCLUSION These RT and chemotherapy dose-response relationships can better inform contemporary RT planning for pediatric patients and surveillance guidelines for high-risk survivors.

ACCOMPANYING CONTENT

Data Supplement

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CONTEXT

Key Objective

Do chemotherapy (ie, cumulative dose) and radiotherapy (ie, mean, max, and dose-volume metrics specifically to the colorectum) dose-response relationships for the incidence of colorectal subsequent malignant neoplasms (SMNs) provide better risk stratification compared with less specific treatment descriptors (eg, yes *v* no exposure)?

Knowledge Generated

Novel and valuable to pediatric radiation oncology is our finding that colorectum SMN risk increases when the colorectum receives high mean radiation therapy doses ≥10 Gy and when ≥20% of the volume is irradiated, being most pronounced when large volumes receive high doses.

Relevance *(S. Bhatia)*

These radiation and chemotherapy dose-response relationships can better inform contemporary radiation planning for future patients with childhood cancer, as well as risk-stratified surveillance guidelines for childhood cancer survivors.*

*Relevance section written by *JCO* Associate Editor Smita Bhatia, MD, MPH, FASCO.

INTRODUCTION

Subsequent malignant neoplasms (SMNs) are the second leading cause of death in childhood cancer survivors.[1-4] The risk for a subsequent colorectal cancer is three- to four-fold higher in childhood cancer survivors compared with the general population.[5,6] Both radiation therapy (RT) and chemotherapy are established risk factors for colorectal SMNs in childhood cancer survivors.[6-15] However, previous studies evaluating chemotherapy and colorectal SMN risk in childhood cancer survivors were limited by small sample size, and RT studies have not considered RT doses specifically to the colorectum. For previously explored RT variables, categorical classification[9-12] (RT as yes/no), SMN proximity categories,[6,9] RT delivered to the abdomen and/or pelvis,[11,13] and dose to the SMN site[7,8,14] have been reported as risk factors for colorectal SMNs (or SMN studies that include colorectal cases) in childhood cancer survivors. Although these less specific RT metrics are valid descriptors, they do not take patient-specific RT field blocking into account and thus do not fully capture dose distributions across the colorectum or to specific colorectal segments.[16] A recent case-control study[17] among survivors of primarily adult-onset Hodgkin lymphoma reported a linear dose-response between colorectal-specific mean RT dose and colorectal SMNs, which was stronger with increasing cumulative procarbazine dose. Yet, there remains a paucity of data regarding RT metrics commonly used by the radiation oncology community that are applicable to patients with childhood cancer and survivors. Specifically, RT mean dose and dose-volume to the colorectum and its segments have not been investigated.

Colorectal cancer screening has reduced mortality from colorectal cancer in the general population.[18-22] Thus, it is critical to identify treatment-related risk factors associated with colorectal SMNs to inform screening strategies in high-risk survivors. Additionally, for newly diagnosed children requiring abdomen and/or pelvic RT, establishing risk associated with dose-volume metrics could inform treatment plan optimization and/or selection of alternate treatment modalities (eg, proton *v* photon) to reduce colorectum dose. Therefore, the objectives of this study were to evaluate the associations between colorectal SMN incidence and colorectal-specific RT with dose-volume data and chemotherapeutic doses in a large childhood cancer survivor cohort.

METHODS

Population

The Childhood Cancer Survivor Study (CCSS) is a retrospective cohort with longitudinal follow-up of 25,723 five-year survivors of childhood cancer diagnosed between 1970 and 1999 with a variety of malignancies before age 21 years at one of 31 North American institutions. Cohort details were previously published.[23-25]

SMN Ascertainment

Within the CCSS cohort, 104 colorectal SMNs were identified from self-report questionnaires or the National Death Index and confirmed by pathology report, medical record, or death certificate review. The anatomic site and pathologic subtype of colorectal malignancies were based on the International Classification of Diseases for Oncology, Third Edition (ICD-O-3), with a fifth digit behavior code of 3.[26] For tumor location, ICD-O-3 site codes were categorized as colon-only or rectum-only and by colorectal subsites for analyses (Data Supplement, Table S1, online only).

TABLE 1. Demographic and Treatment Characteristics of Survivors With and Without Colorectal SMN Diagnosis

Characteristic	Survivors With Colorectal SMN (n = 104)	Survivors Without Colorectal SMN (n = 25,619)
Age at primary cancer diagnosis of childhood cancer, years, median (IQR)	10.0 (3.6-14.9)	6.1 (3.0-12.4)
Age at last follow-up (years), median (IQR)	45.2 (36.9-51.3)	36.0 (28.7-43.8)
Duration of follow-up from primary diagnosis to last follow-up (years), median (IQR)	34.3 (29.8-41.1)	28.5 (22.2-35.4)
RT-specific dose characteristics,[b] median (IQR)		
Colorectal mean RT dose[c] (Gy)	9.4 (1.0-16.0)	0.4 (0.1-7.0)
Colorectal maximum RT dose[d] (Gy)	23.2 (4.5-40.3)	2.2 (0.1-23.0)
$V_{5\ Gy}$[e] (%)	47.5 (32.5-100.0)	34.0 (15.0-70.0)
$V_{20\ Gy}$[e] (%)	35.0 (26.0-58.0)	19.0 (11.0-33.0)
$V_{30\ Gy}$[e] (%)	30.0 (24.0-58.0)	17.0 (9.0-31.0)
$V_{40\ Gy}$[e] (%)	14.0 (6.0-30.0)	16.0 (8.0-30.0)
Sex, No.[a] (%)		
Female	41 (40.9)	11,926 (46.5)
Male	63 (59.1)	13,693 (53.5)
Race, No.[a] (%)		
White, non-Hispanic	87 (84.1)	20,409 (78.8)
Black, non-Hispanic	5 (4.7)	1,629 (6.4)
Hispanic	5 (4.7)	2,055 (8.9)
Other	7 (6.6)	1,526 (5.8)
Age at primary cancer diagnosis, years, No.[a] (%)		
<1	3 (2.8)	1,935 (6.6)
1-3	25 (25.9)	6,396 (28.2)
4-7	14 (13.1)	5,701 (24.0)
8-10	12 (11.3)	2,897 (11.5)
11-14	27 (25.3)	6,223 (21.4)
15-20	23 (21.6)	2,467 (8.3)
Age at the end of follow-up, years, No.[a] (%)		
0-19	2 (1.9)	1,134 (4.3)
20-29	17 (15.9)	5,164 (24.9)
30-39	33 (33.4)	8,738 (34.9)
40-49	38 (35.6)	7,195 (24.5)
50+	14 (13.1)	3,388 (11.4)
Vital status, No.[a] (%)		
Alive	71 (69.1)	21,546 (85.7)
Dead	33 (30.9)	4,075 (14.3)
Primary diagnosis, No.[a] (%)		
Leukemia	11 (12.8)	7,897 (40.3)
CNS tumor	11 (10.3)	4,486 (15.1)
Hodgkin lymphoma	37 (34.7)	3,074 (10.4)
Wilms tumor	13 (12.2)	2,264 (7.6)
Non-Hodgkin lymphoma	12 (11.3)	2,105 (7.1)
Bone tumor	9 (8.4)	2,091 (7.0)
Neuroblastoma	3 (2.8)	1,951 (6.6)
Soft tissue sarcoma	8 (7.5)	1,752 (5.9)
Treatment era, No.[a] (%)		
1970-1979	49 (46.0)	6,557 (22.1)
1980-1989	45 (44.7)	10,010 (36.7)
1990-1999	10 (9.4)	9,052 (41.2)

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TABLE 1. Demographic and Treatment Characteristics of Survivors With and Without Colorectal SMN Diagnosis (continued)

Characteristic	Survivors With Colorectal SMN (n = 104)	Survivors Without Colorectal SMN (n = 25,619)
Radiation therapy, No.[a] (%)		
Yes	75 (78.7)	13,073 (51.1)
No	21 (21.3)	10,436 (48.9)
Chemotherapy, No.[a] (%)		
Yes	86 (89.9)	19,072 (84.0)
No	10 (10.1)	4,379 (16.0)
Alkylating agents, No.[a] (%)		
Yes	69 (72.6)	12,722 (54.9)
No	27 (27.4)	10,720 (45.1)
Platinum agents, No.[a] (%)		
Yes	11 (11.2)	2,755 (10.0)
No	85 (88.8)	20,784 (90.0)
Anthracyclines, No.[a] (%)		
Yes	46 (49.3)	11,404 (53.3)
No	50 (50.7)	12,088 (46.7)
Procarbazine, No.[a] (%)		
Yes	35 (35.5)	2,123 (7.7)
No	61 (64.5)	21,403 (92.3)
Time from primary cancer diagnosis to colorectal SMN diagnosis, years, median (IQR)	29.9 (24.0-35.4)	NA
Age at colorectal SMN diagnosis, years, median (IQR)	39.7 (31.8-45.2)	NA
Age at colorectal SMN, years, No.[a] (%)		
15 to <20	2 (2.0)	NA
20 to <25	9 (9.0)	NA
25 to <30	8 (8.0)	NA
30 to <35	14 (13.0)	NA
35 to <40	19 (18.0)	NA
40 to <45	22 (21.0)	NA
45+	30 (29.0)	NA
Pathologic subtype of colorectal SMN, No.[a] (%)		
Adenocarcinoma, NOS	77 (74.7)	NA
Mucinous adenocarcinoma	8 (7.5)	NA
Neoplasm, malignant	5 (4.7)	NA
Adenocarcinoma in adenomatous polyp	4 (3.8)	NA
Carcinoid tumor, NOS	2 (1.9)	NA
Carcinoma, NOS	2 (1.9)	NA
Neuroendocrine carcinoma, NOS	2 (1.9)	NA
Adenocarcinoma in callous adenoma	1 (0.9)	NA
Goblet cell carcinoid	1 (0.9)	NA
Medullary carcinoma, NOS	1 (0.9)	NA
Signet ring cell carcinoma	1 (0.9)	NA
Anatomic site of SMN, No.[a] (%)[f]		
Cecum	7 (6.6)	NA
Appendix	4 (3.8)	NA
Ascending colon	12 (11.3)	NA
Hepatic flexure	1 (0.9)	NA
Transverse colon	7 (6.6)	NA
Splenic flexure	4 (3.8)	NA
Descending colon	6 (5.6)	NA
Sigmoid colon	15 (14.1)	NA

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TABLE 1. Demographic and Treatment Characteristics of Survivors With and Without Colorectal SMN Diagnosis (continued)

Characteristic	Survivors With Colorectal SMN (n = 104)	Survivors Without Colorectal SMN (n = 25,619)
Large intestine, NOS	10 (9.4)	NA
Rectosigmoid junction	14 (15.6)	NA
Rectum	24 (22.5)	NA
Cause of death of colorectal SMN participants, No.[a] (%)		
Alive	71 (69.1)	NA
Colorectal SMN	20 (18.8)	NA
External causes (eg, accident, injury)	7 (6.6)	NA
Cardiac toxicity	3 (2.8)	NA
Other SMN	2 (1.9)	NA
Primary cancer	1 (0.9)	NA

NOTE. Data are median (IQR) or No. (%). Percentages are reported among participants with known values.
Abbreviations: NA, not applicable; NOS, not otherwise specified; RT, radiation therapy; SMN, subsequent malignant neoplasm.
[a]To account for planned undersampling of ALL in the expansion cohort (1987-1999), all analyses except the reported frequency counts were weighted. Percentages were based on available data.
[b]Data on the basis of survivors treated with RT and with known values for each respective radiation variable.
[c]Colorectal mean RT dose is the mean dose to the colorectum.
[d]Colorectal maximum RT dose is the maximum RT dose to the colorectum.
[e]$V_{5\ Gy}$, $V_{10\ Gy}$, $V_{20\ Gy}$, $V_{30\ Gy}$, and $V_{40\ Gy}$ are the percent volume receiving dose ≥5, 10, 20, 30, and 40 Gy, respectively.
[f]Anatomic site of colorectal SMN on the basis of site recode definitions from the *International Classification of Diseases for Oncology, Third Edition* (ICD-O-3).

Treatment Exposures

Demographic and primary cancer-related characteristics were collected from questionnaire responses.[25] Patient records were reviewed for cancer treatment exposures.[25] Treatment exposures included as yes/no qualitative variables were chemotherapy, chemotherapeutic group types, procarbazine, nonprocarbazine alkylating agents,[15] stem-cell transplant, RT, RT to specific body regions, and total-body irradiation. Chemotherapy cumulative doses included doxorubicin-equivalent dose[27] for anthracyclines, cyclophosphamide-equivalent dose[28] for alkylating agents, and platinum cumulative dose[29,30] for platinum agents. We included procarbazine because previous studies reported increased colorectal SMN risk with this specific alkylating agent.[6,17,31] For nonprocarbazine alkylating agents, procarbazine exposure/dose was removed from the cumulative alkylating agent dose (Data Supplement, Table S2, contains a list of specific agents and their frequency of use). RT dose variables included the maximum target RT dose to the abdomen or pelvis, colorectal maximum RT dose, mean colorectal RT dose (mean colorectal dose [MCD]), and mean RT dose to each colorectal segment. Dose-volume data consisted of the percent of colorectal volume receiving ≥5($V_{5\ Gy}$), 10($V_{10\ Gy}$), 20($V_{20\ Gy}$), 30($V_{30\ Gy}$), and 40($V_{40\ Gy}$) Gy. We included $V_{5\ Gy}$ and $V_{10\ Gy}$ with maximum dose <20 Gy ($V_{20\ Gy\ =\ 0\%}$) to evaluate the effect of low doses to large volumes. Our validated colorectum RT dosimetry methods were previously published[16,32-34] and are briefly summarized in the Data Supplement. Dose-volume metrics are visually illustrated in the Data Supplement (Fig S1).

Statistical Analysis

We calculated standardized incidence ratios (SIRs) and absolute excess risks (AERs) for colorectal SMN incidence for all CCSS participants compared with expected number in the general population from SEER.[35] Demographics, health behaviors, and treatment exposures were considered. Cumulative incidences of colorectal SMN overall and by subgroup were estimated using years since primary childhood cancer diagnosis accounting for death as the competing risk; incidences were also calculated using attained age.[36] Incidence rates by age at colorectal cancer diagnosis were calculated for the overall CCSS cohort and by high-risk groups and were compared with general population rates from SEER.[37]

Multivariable piecewise-exponential models assessed the adjusted associations of the treatment-related risk factors and multiplicative interactions between RT and chemotherapy with colorectal SMN incidence. Adjusted IRRs were calculated from piecewise-exponential models adjusted for several factors on the basis of a priori hypotheses and for significant (P < .05) univariate treatment-related factors (univariate screening details and results are provided in the Data Supplement, Table S3). We assessed collinearity between MCD with each dose-volume metric using correlation coefficients (results and discussion provided in the Data Supplement).

TABLE 2. SIRs, AERs, and Cumulative Incidences for Colorectal SMN Overall and by Subgroup

Characteristic	No. Observed	No. Observed[a] (considering weights)	No. Expected	SIR (95% CI)	AER/100,000 PY (95% CI)[e]	40-Year Cumulative Incidence (95% CI; years since Dx as time scale)	50-Year Cumulative Incidence (95% CI; attained age as time scale)
All survivors	104	106.6	40.3	2.6 (2.2 to 3.2)	9.4 (6.7 to 12.5)	0.7 (0.5 to 0.8)	0.7 (0.6 to 0.9)
Site of colorectal SMN[b]							
Colon only	70	72.6	26.7	2.7 (2.1 to 3.4)	6.5 (4.3 to 9.2)	0.5 (0.4 to 0.6)	0.5 (0.3 to 0.6)
Rectum only	24	24.0	12.4	1.9 (1.2 to 2.9)	1.6 (0.4 to 3.3)	0.2 (0.1 to 0.3)	0.2 (0.1 to 0.3)
Age at primary cancer diagnosis, years							
<1	3	3.0	1.0	2.9 (0.6 to 8.5)	4.0 (−0.9 to 15.6)	0.2 (0.0 to 0.4)	0.6 (0.0 to 1.4)
1-3	25	27.6	4.6	6.0 (4.0 to 8.7)	11.6 (6.9 to 17.9)	0.8 (0.4 to 1.1)	1.1 (0.5 to 1.7)
4-7	14	14.0	6.2	2.3 (1.2 to 3.8)	4.6 (0.9 to 10.2)	0.4 (0.2 to 0.7)	0.5 (0.2 to 0.9)
8-10	12	12.0	4.6	2.6 (1.3 to 4.5)	9.2 (2.0 to 20.5)	0.8 (0.3 to 1.3)	0.8 (0.3 to 1.4)
11-14	27	27.0	11.8	2.3 (1.5 to 3.3)	10.8 (4.3 to 19.6)	1.0 (0.5 to 1.4)	0.7 (0.4 to 1.0)
15-20	23	23.0	12.1	1.9 (1.2 to 2.8)	15.8 (3.6 to 32.5)	0.7 (0.4 to 1.1)	0.6 (0.3 to 0.9)
Attained age, years							
5-14	0	0.0	0.1	0.0 (0.0 to 73.2)	−0.0 (−0.0 to 3.3)	NA	NA
15-24	11	11.0	1.6	6.9 (3.5 to 12.4)	4.0 (1.7 to 7.7)	NA	NA
25-34	22	24.6	7.0	3.5 (2.3 to 5.2)	8.4 (4.2 to 14.0)	NA	NA
35-44	41	41.0	13.5	3.0 (2.2 to 4.1)	25.2 (14.6 to 38.6)	NA	NA
45-54	25	25.0	14.8	1.7 (1.1 to 2.5)	28.4 (3.8 to 61.6)	NA	NA
55+	5	5.0	3.3	1.5 (0.5 to 3.5)	34.4 (−35.0 to 170.9)	NA	NA
Years since primary cancer diagnosis							
5-14	6	6.0	1.6	3.7 (1.3 to 8.0)	1.5 (0.2 to 4.0)	NA	NA
15-24	23	23.0	8.0	2.9 (1.8 to 4.3)	6.1 (2.7 to 10.7)	NA	NA
25-34	47	49.6	16.6	3.0 (2.2 to 3.9)	24.9 (15.2 to 36.8)	NA	NA
35+	28	28.0	14.0	2.0 (1.3 to 2.9)	35.0 (11.4 to 66.2)	NA	NA
Sex							
Male	63	63.0	22.6	2.8 (2.1 to 3.6)	10.8 (6.9 to 15.5)	0.8 (0.5 to 1.0)	0.9 (0.6 to 1.1)
Female	41	43.6	17.7	2.5 (1.8 to 3.3)	7.8 (4.2 to 12.3)	0.6 (0.4 to 0.8)	0.6 (0.4 to 0.8)
Race/ethnicity							
White, non-Hispanic	87	89.6	34.0	2.6 (2.1 to 3.2)	9.8 (6.7 to 13.5)	0.7 (0.5 to 0.8)	0.7 (0.6 to 0.9)
Black, non-Hispanic	5	5.0	1.7	2.9 (0.9 to 6.7)	7.9 (−0.3 to 24.0)	0.8 (0.0 to 1.7)	0.6 (0.0 to 1.2)
Hispanic	5	5.0	2.2	2.3 (0.7 to 5.3)	5.0 (−1.1 to 16.8)	0.6 (0.0 to 1.2)	0.6 (0.0 to 1.2)
Other	7	7.0	2.4	2.9 (1.2 to 6.0)	10.7 (0.9 to 28.0)	0.6 (0.0 to 1.1)	0.7 (0.1 to 1.4)
First-degree relative with colorectal malignancy							
Yes	3	3.0	0.5	6.1 (1.2 to 17.7)	42.9 (1.8 to 141.7)	0.9 (0.0 to 2.2)	1.3 (0.0 to 3.0)
No	101	103.6	39.8	2.6 (2.1 to 3.2)	9.1 (6.4 to 12.2)	0.7 (0.5 to 0.8)	0.7 (0.6 to 0.9)

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TABLE 2. SIRs, AERs, and Cumulative Incidences for Colorectal SMN Overall and by Subgroup (continued)

Characteristic	No. Observed	No. Observed[a] (considering weights)	No. Expected	SIR (95% CI)	AER/100,000 PY (95% CI)[e]	40-Year Cumulative Incidence (95% CI; years since Dx as time scale)	50-Year Cumulative Incidence (95% CI; attained age as time scale)
Treatment era							
1970s	49	49.0	22.0	2.2 (1.7 to 2.9)	12.7 (6.7 to 20.2)	0.6 (0.4 to 0.8)	0.7 (0.5 to 0.9)
1980s	45	47.6	14.1	3.4 (2.5 to 4.5)	11.9 (7.4 to 17.4)	0.7 (0.4 to 0.9)	0.8 (0.5 to 1.2)
1990s	10	10.0	4.3	2.3 (1.1 to 4.3)	2.7 (0.2 to 6.7)	0.2 (0.0 to 0.3)[d]	0.2 (0.0 to 0.4)
Primary diagnosis							
Leukemia	11	13.6	11.2	1.2 (0.7 to 2.1)	0.9 (−1.4 to 4.3)	0.2 (0.1 to 0.4)	0.4 (0.1 to 0.6)
CNS	11	11.0	4.8	2.3 (1.1 to 4.1)	6.4 (0.7 to 15.2)	0.5 (0.1 to 0.9)	0.6 (0.1 to 1.0)
Hodgkin lymphoma	37	37.0	8.4	4.4 (3.6 to 11.7)	37.2 (23.0 to 55.4)	1.6 (1.0 to 2.2)	1.3 (0.8 to 1.8)
Non-Hodgkin lymphoma	12	12.0	3.9	3.1 (1.6 to 5.4)	15.5 (4.4 to 32.7)	1.0 (0.3 to 1.6)	0.9 (0.3 to 1.6)
Wilms tumor	13	13.0	1.9	6.8 (3.6 to 11.7)	19.0 (8.6 to 34.7)	1.5 (0.6 to 2.4)	1.7 (0.5 to 2.8)
Neuroblastoma	3	3.0	1.2	2.4 (0.5 to 7.1)	3.7 (−1.3 to 15.6)	0.3 (0.0 to 0.7)	0.7 (0.0 to 1.8)
Soft tissue sarcoma	8	8.0	3.8	2.1 (0.9 to 4.2)	9.0 (−0.7 to 25.5)	0.6 (0.2 to 1.1)	0.5 (0.1 to 1.0)
Bone cancer	9	9.0	5.2	1.7 (0.8 to 3.3)	7.6 (−2.2 to 23.8)	0.6 (0.2 to 1.0)	0.6 (0.2 to 1.0)
History of any RT							
Yes	75	77.6	23.5	3.3 (2.6 to 4.1)	15.4 (10.8 to 21.0)	0.8 (0.6 to 1.0)	0.8 (0.6 to 1.0)
No	21	21.0	12.8	1.6 (1.0 to 2.5)	2.7 (0.0 to 6.5)	0.4 (0.2 to 0.6)	0.5 (0.2 to 0.7)
History of abdominal RT							
Yes	47	47.0	9.2	5.1 (3.8 to 6.8)	27.7 (18.6 to 39.1)	1.2 (0.8 to 1.6)	1.1 (0.8 to 1.5)
No	45	47.6	26.3	1.8 (1.3 to 2.4)	4.3 (1.8 to 7.4)	0.5 (0.3 to 0.6)	0.5 (0.3 to 0.7)
History of pelvis RT							
Yes	43	43.0	7.5	5.7 (4.1 to 7.7)	31.3 (20.8 to 44.5)	1.3 (0.8 to 1.7)	1.2 (0.8 to 1.6)
No	49	51.6	28.0	1.8 (1.4 to 2.4)	4.5 (2.0 to 7.7)	0.5 (0.3 to 0.7)	0.5 (0.4 to 0.7)
History of abdominal or pelvis RT							
Yes	53	53.0	10.0	5.3 (4.0 to 6.9)	29.4 (20.3 to 40.5)	1.3 (0.9 to 1.7)	1.2 (0.8 to 1.6)
No	39	41.6	25.5	1.6 (1.2 to 2.2)	3.3 (0.9 to 6.3)	0.4 (0.3 to 0.5)	0.5 (0.3 to 0.7)
Colorectum mean RT dose (Gy)[c]							
None	21	21.0	12.8	1.6 (1.0 to 2.5)	2.7 (0.0 to 6.5)	0.4 (0.2 to 0.6)	0.5 (0.2 to 0.7)
0.1 to <10	36	38.6	18.0	2.1 (1.5 to 2.9)	7.6 (3.4 to 12.8)	0.6 (0.4 to 0.8)	0.5 (0.3 to 0.7)
10 to <20	21	21.0	3.4	6.2 (3.8 to 9.4)	36.3 (19.8 to 59.2)	1.5 (0.8 to 2.2)	1.6 (0.8 to 2.3)
≥20	13	13.0	1.0	12.6 (6.7 to 21.5)	101.1 (49.7 to 179.1)	2.8 (1.1 to 4.6)	3.1 (1.2 to 4.8)
History of alkylating agents							
Yes	69	71.6	18.8	3.8 (3.0 to 4.8)	15.5 (10.9 to 21.0)	0.9 (0.6 to 1.2)	0.9 (0.7 to 1.2)
No	27	27.0	17.3	1.6 (1.0 to 2.3)	3.2 (0.2 to 7.2)	0.5 (0.3 to 0.7)	0.5 (0.3 to 0.8)

(continued on following page)

TABLE 2. SIRs, AERs, and Cumulative Incidences for Colorectal SMN Overall and by Subgroup (continued)

Characteristic	No. Observed	No. Observed[a] (considering weights)	No. Expected	SIR (95% CI)	AER/100,000 PY (95% CI)[e]	40-Year Cumulative Incidence (95% CI; years since Dx as time scale)	50-Year Cumulative Incidence (95% CI; attained age as time scale)
History of platinum agents							
Yes	11	11.0	1.9	5.9 (3.0 to 10.6)	17.8 (7.1 to 34.8)	0.9 (0.2 to 1.5)	0.7 (0.2 to 1.1)
No	85	87.6	34.4	2.5 (2.0 to 3.1)	8.9 (6.0 to 12.3)	0.7 (0.5 to 0.9)	0.7 (0.5 to 0.9)
History of anthracyclines							
Yes	46	48.6	14.8	3.3 (2.4 to 4.4)	10.7 (6.7 to 15.7)	1.1 (0.0 to 1.5)	1.0 (0.6 to 1.3)
No	50	50.0	21.2	2.3 (1.7 to 3.1)	8.7 (4.8 to 13.5)	0.5 (0.0 to 0.7)	0.6 (0.4 to 0.8)
History of procarbazine							
Yes	35	35.0	4.4	7.9 (5.5 to 11.0)	61.6 (40.2 to 89.2)	2.4 (1.4 to 3.3)	1.8 (1.1 to 2.5)
No	61	63.6	31.7	2.0 (1.5 to 2.6)	5.3 (2.9 to 8.3)	0.5 (0.4 to 0.7)	0.6 (0.4 to 0.8)

NOTE. To account for planned undersampling of ALL in the expansion cohort (1987-1999), all analyses except the reported frequency counts were weighted.

Abbreviations: AER, absolute excess risk; CCSS, Childhood Cancer Survivor Study; Dx, diagnosis; NA, not applicable; PY, person-year; RT, radiation therapy; SIR, standardized incidence ratio; SMN, subsequent malignant neoplasm.

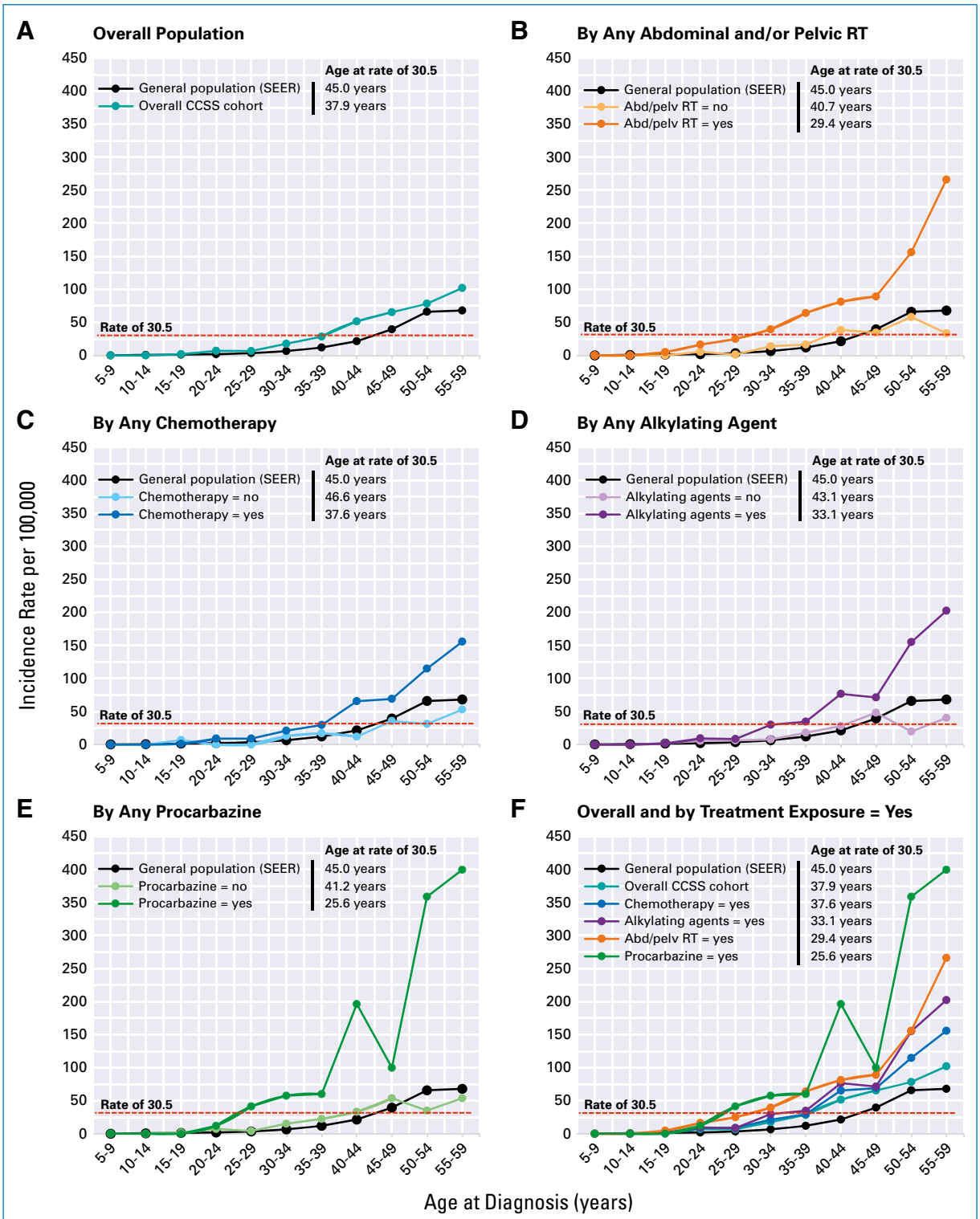
[a]Observed values are reported with decimal places as these values considered weights.

[b]This does not include all colorectal SMNs since 10 cases could not be sorted into the colon or rectum category because of unspecified site.

[c]Colorectal mean RT dose is the mean dose to the colorectum.

[d]In 2020, these survivors were maximally 30 years after primary childhood cancer diagnosis.

[e]AERs were calculated as the difference between the numbers of observed and expected events divided by the number of person-years of follow-up.



G **No. of Colorectal Cancer Cases**

	No. of colorectal cancer cases										
Age at Diagnosis (years)	5-9	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59
General US population (SEER data)	70	342	696	1,048	1,891	3,560	6,289	10,322	18,422	31,144	32,745
Overall CCSS population[a]	0	0	2	9	8	16.64	19	22	16	9	5

FIG 1. Incidence rates of colorectal cancer per 100,000. (A) Rates of colorectal cancer by age at diagnosis for the general US population and for the CCSS cohort (all cancer treatments combined). For the CCSS cohort, rates were also stratified by (B) any abdominal and/or pelvic RT, (C) any chemotherapy, (D) any alkylating agent, and (E) any procarbazine. (F) The same data—data from subplot A and data from the yes category for each of the treatment variables from (continued on following page)

FIG 1. (Continued). subplots (B-E)—were plotted together to better visualize these data on the same plot. The rates for the general US population were calculated using SEER incidence data (November 2024 submission, SEER 21 registries) from 2018 to 2022. In the United States, the colorectal cancer screening guideline for the general population is to start at age 45 years, and the incidence rates of colorectal cancer at 40-44 years and 45-49 years are 21.5 and 39.5 per 100,000, respectively. Using linear interpolation, the incidence at the recommended screening age is 30.5 per 100,000, which is depicted by the dashed red line in subplots (A-F). For the overall CCSS cohort and each of the treatment-related risk groups, the age (also calculated using linear interpolation) at which the rate of 30.5 occurs is shown in the table at the upper right-hand corner of each subplot. The number of colorectal cancer cases in each of the populations by age at diagnosis is shown in (G). Exact values and their corresponding 95% confidence intervals are displayed in the Data Supplement (Table S5). [a]The count of colorectal cancer cases are reported with decimal places as these values considered weights. CCSS, Childhood Cancer Survivor Study; RT, radiation therapy.

Subanalyses restricted the analysis cohort as follows: for chemotherapy variables, we reported multivariable models restricted to survivors not treated with RT to evaluate chemotherapy effects while fully excluding the potential residual confounding effect of RT. For nonprocarbazine alkylating agents, we excluded survivors treated with procarbazine. Similarly, for RT variables, we excluded survivors treated with procarbazine.

For MCD and procarbazine dose, we reported adjusted excess rate ratio (ERR) models to quantify the excess risk of colorectal SMN associated with each unit increase of dose (ie, gray for RT dose and mg/m^2 for procarbazine doses). ERR models were adjusted for the same variables used in the adjusted piecewise-exponential models. We compared ERR models with the linear RT form and quadratic RT form using a likelihood ratio test.[38] We also examined if there is an additive interaction between MCD and procarbazine dose.

Interactions between MCD and procarbazine were tested in the piecewise-exponential model and the ERR model using the likelihood ratio test. *P* values for interaction were adjusted for multiple testing of seven interactions by Bonferroni correction. All reported *P* values were two-sided and considered statistically significant at the .05 level. All statistical analyses were performed using SAS (version 9.4; SAS Institute, Cary, NC).

RESULTS

Cohort Characteristics

Table 1 provides demographic and treatment characteristics of the survivors (N = 25,723) as well as clinical and pathologic characteristics of the colorectal SMN cases (n = 104). The median time from primary cancer diagnosis to colorectal SMN diagnosis was 29.9 years (IQR, 24.0-35.4). The median age at colorectal SMN diagnosis was 39.7 years (IQR, 31.8-45.2). Nineteen (18%) cases occurred in individuals younger than 30 years. Thirty-three patients with a colorectal SMN were deceased, of whom 20 (60.6%) died of colorectal SMN.

SIRs and AERs

The SIR for colorectal SMN was 2.6-fold higher in childhood cancer survivors than in the general population with 9.4 additional colorectal SMN cases per 100,000 person-years (Table 2). SIRs were elevated for survivors who received anthracyclines (SIR, 3.3 [95% CI, 2.4 to 4.4]), alkylators (SIR, 3.8 [95% CI, 3.0 to 4.8]), abdominal and/or pelvic RT (SIR, 5.3 [95% CI, 4.0 to 6.9]), platinum-based agents (SIR, 5.9 [95% CI, 3.0 to 10.6]), and procarbazine (SIR, 7.9 [95% CI, 5.5 to 11.0]). Among treatment-related variables, the AER was highest (61.6 per 100,000 person-years) for those exposed to procarbazine.

Cumulative Incidences

The cumulative incidence of colorectal SMN by 40 years after the primary cancer diagnosis was 0.7% (95% CI, 0.5% to 0.8%) for the overall cohort. Table 2 describes cumulative incidence by various treatment-related factors. Notably, for MCD, alkylating dose, and procarbazine dose, the 40-year cumulative incidence increased with increasing dose. Cumulative incidence by attained age is also provided in Table 2. The Data Supplement (Table S4) presents the full list of cumulative incidences by overall cohort, primary cancer diagnosis, treatment era, and treatment-related variables. Cumulative incidence data were plotted for key treatment-related variables (Data Supplement, Figs S2 and S3).

Incidence Rates per 100,000

For the general US population, the recommended age to initiate colorectal cancer screening is 45 years[39] which corresponds to an incidence rate of 30.5 per 100,000 using SEER data.[37] This same incidence rate was observed at age 37.9 years for the overall CCSS cohort and age 37.6, 33.1, 29.4, and 25.6 years for survivors who received any chemotherapy, alkylating agent, abdominal and/or pelvic RT, or procarbazine, respectively (Fig 1; Data Supplement, Table S5).

Chemotherapy Effects: Piecewise-Exponential Modeling

In multivariable models for chemotherapy associations (Fig 2; Data Supplement, Table S6), rates increased for survivors who received any chemotherapy (IRR, 2.2 [95% CI, 1.1 to 4.6]), anthracycline (IRR, 1.7 [95% CI, 1.1 to 2.6]), alkylating agent (IRR, 2.7 [95% CI, 1.6 to 4.4]), platinum agent (IRR, 3.2 [95% CI, 1.6 to 6.1]), or procarbazine (IRR, 5.5 [95% CI, 3.1 to 9.9]) compared with those not exposed.

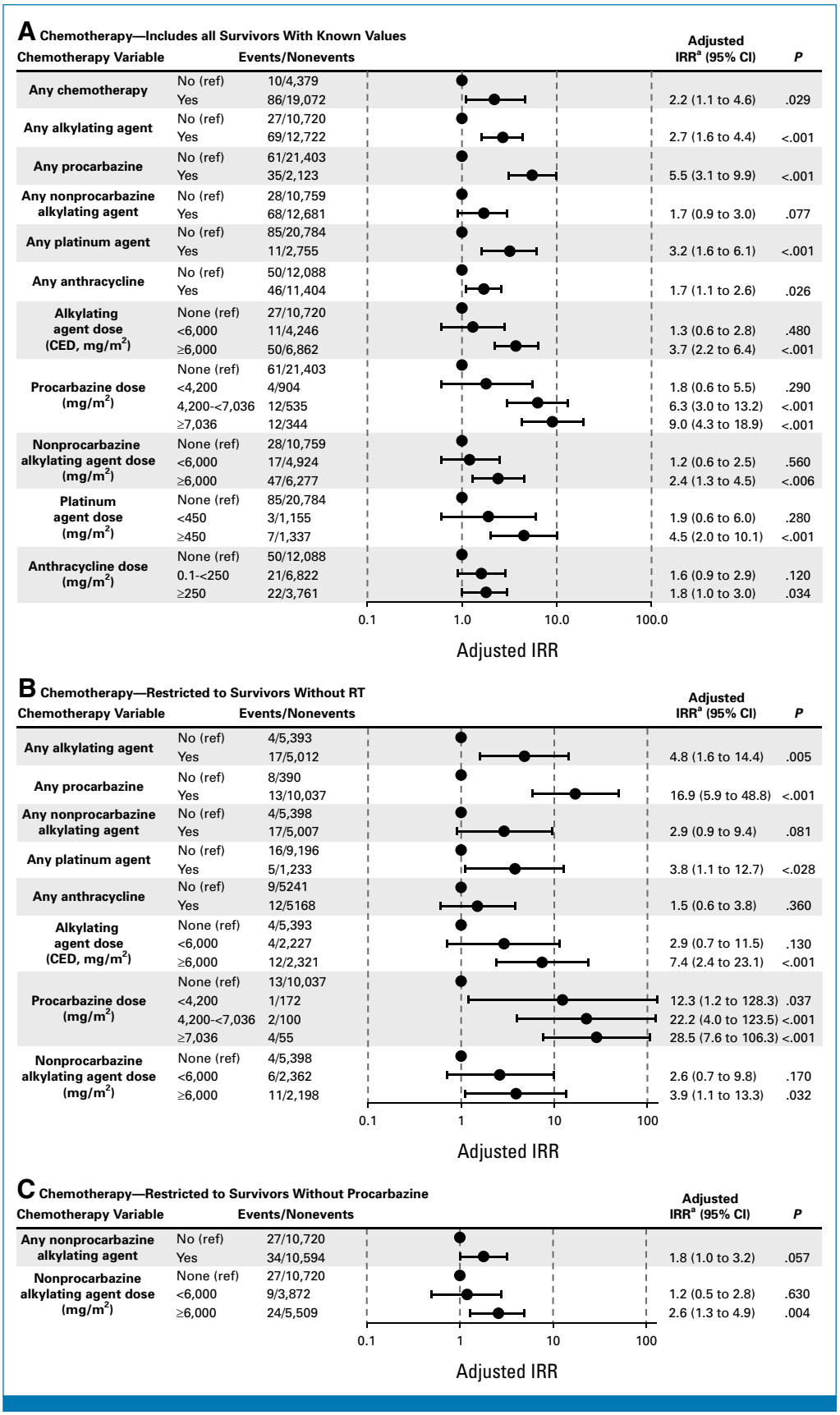


FIG 2. Multivariable-adjusted IRRs for chemotherapy exposures. IRRs by chemotherapy variables (A) including all survivors with known values for treatment-related variables and (B) restricted to survivors without RT to fully exclude a residual confounding effect of RT. (C) IRRs for (continued on following page)

FIG 2. (Continued). nonprocarbazine alkylating agents were also restricted to survivors without procarbazine to fully exclude a residual confounding effect of procarbazine. IRRs were calculated from piecewise-exponential models. The reference group for all models consisted of survivors who did not receive any of the chemotherapeutic variable being assessed. The *x*-axis is in logarithmic scale and displays the IRR. Adjusted models were adjusted for attained age, age at primary cancer diagnosis, sex, and race/ethnicity on the basis of a priori hypotheses. On the basis of univariate significance ($P < .1$), chemotherapy-specific models were also adjusted for mean colorectal RT dose (cutoffs: none, 0.1 to <10, 10 to <20, and ≥20 Gy). Nonprocarbazine alkylating agents models were additionally adjusted for procarbazine dose (cutoffs: 0, <4,200, 4,200 to <7,036, and ≥7,036 mg/m^2) since procarbazine and nonprocarbazine alkylating agents are highly correlated. Individuals with missing data relevant to each model were excluded. [a]Each row came from a separate model. CED, cyclophosphamide equivalent dose; events, survivors with a colorectal SMN; IRR, incidence rate ratio; nonevents, survivors without a colorectal SMN; ref, reference; RT, radiation therapy; SMN, subsequent malignant neoplasm.

For dose-response analysis, increased rates were observed for doxorubicin-equivalent dose ≥250 mg/m² (IRR, 1.8 [95% CI, 1.0 to 3.0]), cyclophosphamide-equivalent dose ≥6,000 mg/m² (IRR, 3.7 [95% CI, 2.2 to 6.4]), and platinum dose ≥450 mg/m² (IRR, 4.5 [95% CI, 2.0 to 10.1]). For procarbazine, increased rates were observed for cumulative dose ≥4,200 mg/m² with IRRs of 6.3 (95% CI, 3.0 to 13.2) and 9.0 (95% CI, 4.3 to 18.9) for doses of 4,200 to <7,036 and ≥7,036 mg/m², respectively. When restricting chemotherapy multivariable models to survivors without RT, increased IRRs were observed for those exposed to any alkylating agent (IRR, 4.8 [95% CI, 1.6 to 14.4]), procarbazine (IRR, 16.9 [95% CI, 5.9 to 48.8]), or platinum-based agent (IRR, 3.8 [95% CI, 1.1 to 12.7]; Fig 2B; Data Supplement, Table S7). For nonprocarbazine alkylating agents, when restricting the analysis to survivors without procarbazine (Fig 2C), the high-dose category of ≥6,000 mg/m² was significant (IRR, 2.6 [95% CI, 1.3 to 4.9]).

Radiation Effects: Piecewise-Exponential Modeling

In multivariable models, survivors who received maximum colorectal RT doses 10 to <20 Gy and ≥20 Gy had IRRs of 2.6 (95% CI, 1.1 to 6.0) and 3.0 (95% CI, 1.7 to 5.5), respectively (Fig 3A). Colorectal SMN rates increased with increasing mean RT doses ≥10 Gy to the colorectum or individual colorectal segments (all $P < .01$, except sigmoid; Data Supplement, Table S6). Multivariable RT models were also provided for MCD in 5-Gy bins (Data Supplement, Table S8) and restricted to survivors without procarbazine (Fig 3B). For each dose-volume metric (Fig 3C), rates increased with increasing irradiated volume above 20%. For example, for $V_{20\ Gy}$, the IRR was 3.8 (95% CI, 1.9 to 7.6), 4.9 (95% CI, 2.0 to 12.0), and 8.7 (95% CI, 3.5 to 21.6) for irradiated colorectal volumes of 20% to <40%, 40% to <80%, and ≥80%, respectively (Data Supplement, Fig S4A). Likewise, IRRs increased as the dose to a given volume of colorectum increased. For example, when 20%-40% of the colorectum was irradiated, the IRR was 2.8 (95% CI, 1.3 to 5.9), 3.8 (95% CI, 1.9 to 7.6), 4.5 (95% CI, 2.1 to 9.7), and 5.4 (95% CI, 2.0 to 14.3) when doses were ≥5, 20, 30, and 40 Gy, respectively (Data Supplement, Fig S4B). We also assessed the effect of low dose to large volumes by considering $V_{5\ Gy}$ and $V_{10\ Gy}$ with maximum dose <20 Gy ($V_{20\ Gy} = 0\%$) and did not observe a clear trend of colorectal SMN risk, which may be due to limited sample size (Data Supplement, Table S6). To our knowledge, our study is the first to assess colorectal SMN risk for specific colorectal segments (ascending, transverse, descending, and sigmoid colon and rectum) using segment-specific mean RT dose; similar rates were observed across the different segments (results in the Data Supplement, Table S6, and discussion in the Data Supplement). No multiplicative interaction was found between RT and chemotherapy for colorectal SMN incidence after Bonferroni correction (Data Supplement, Table S9).

ERR Modeling

With linear ERR modeling, the colorectal SMN rate increased by 73.0% (95% CI, 26.4% to 119.6%) per 1,000 mg/m² increase in procarbazine dose and by 20.8% (95% CI, 9.0 to 32.5) per 1 Gy increase in MCD (Fig 4). Quadratic ERR models did not improve the fit compared with the corresponding linear models, suggesting no significant departure from linearity for procarbazine ($P = .18$) or for MCD ($P = .58$). In the linear ERR RT dose model, there was no statistically significant additive interaction between MCD and procarbazine when considered as either yes/no ($P = .26$) or categorical dose ($P = .72$) variables.

DISCUSSION

To our knowledge, this is the most extensive study of colorectal SMN risk among survivors of childhood cancer, comprehensively reporting risk associated with cumulative doses of chemotherapeutic agents and the distribution of RT dose across the colorectum. Consistent with our findings, previous studies of childhood cancer survivors have shown associations between gastrointestinal or colorectal SMN risk and alkylating agents,[8,15] platinum drugs,[6] and anthracyclines,[7] but few studies have assessed dose-dependent relationships with anthracyclines,[7] procarbazine,[6,17] and platinum agents.[40] Results of previous studies were not statistically significant[7,40] or were not focused specifically on colorectal SMNs[6] in pediatric

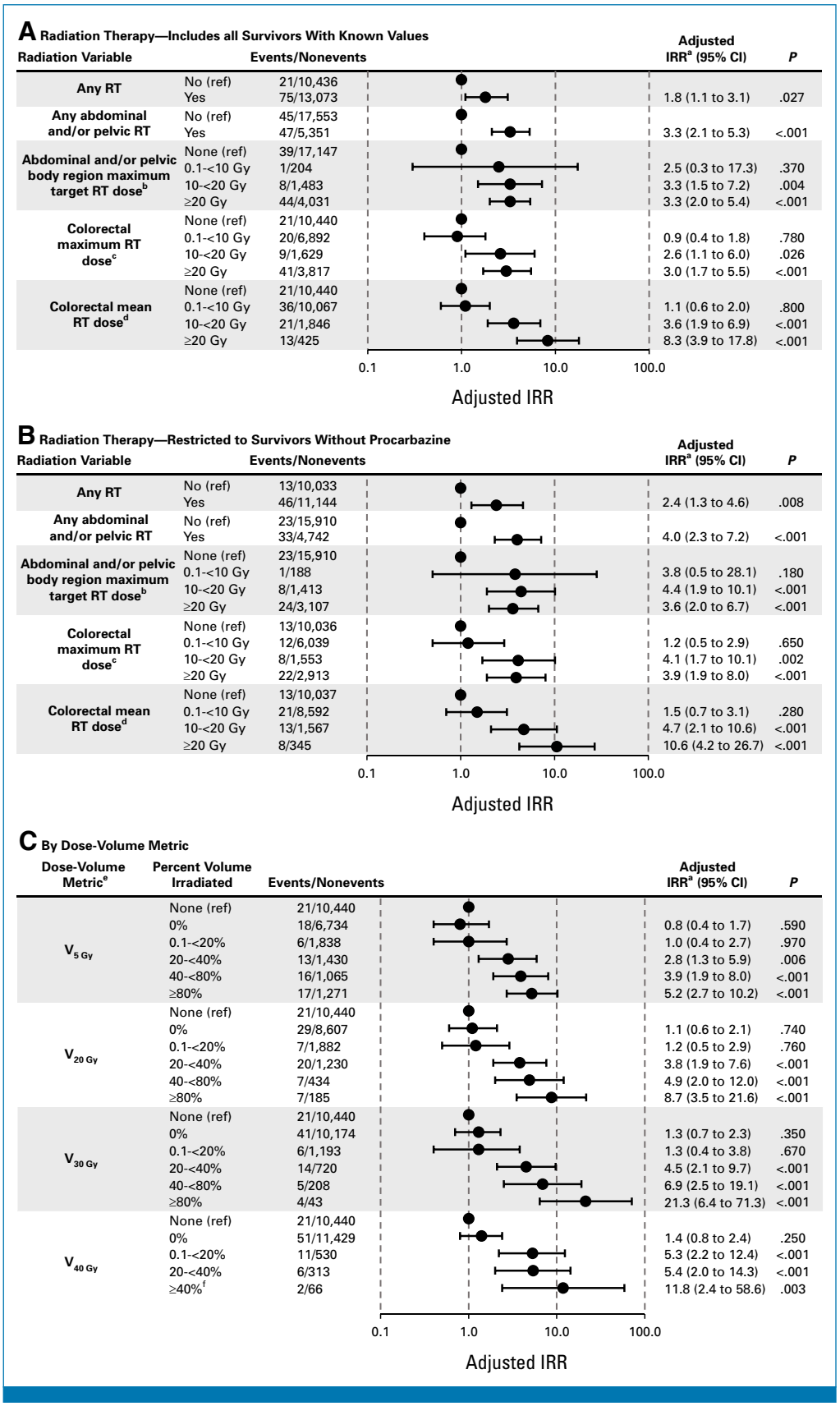


FIG 3. Multivariable-adjusted IRRs for RT exposures. Multivariable IRRs by select RT variables (A) including all survivors with known values for treatment-related variables and (B) restricted to survivors without procarbazine to fully exclude a residual confounding effect of procarbazine. (C) Multivariable IRRs by percent volume of colorectum irradiated for a constant dose-volume metric including (continued on following page)

FIG 3. (Continued). all survivors with known values. See the Data Supplement (Fig S4) for IRRs by dose-volume metric for a constant percent volume irradiated (same data as C but rearranged). IRRs were calculated from piecewise-exponential models. The reference group for all models consisted of survivors who did not receive any RT. The *x*-axis is in logarithmic scale and displays the IRR. Multivariable models were adjusted for attained age, age at primary cancer diagnosis, sex, and race/ethnicity on the basis of a priori hypotheses. On the basis of univariate significance ($P < .1$), RT-specific models were also adjusted for alkylating agents using cyclophosphamide-equivalent dose (cutoffs: 0, <6,000, and ≥6,000 mg/m²). Individuals with missing data relevant to each model were excluded. [a]Each row came from a separate model. [b]The abdominal and/or pelvic body region maximum target RT dose is the maximum target dose to the abdomen or pelvis, which was determined by summing the prescription dose(s) from all overlapping fields in each region. When the abdomen and pelvis regions were treated, the higher maximum target dose was used in our analyses. [c]Colorectal maximum RT dose is the maximum RT dose to the colorectum. [d]Colorectal mean RT dose is the mean dose to the colorectum. [e]$V_{5\ Gy}$, $V_{10\ Gy}$, $V_{20\ Gy}$, $V_{30\ Gy}$, and $V_{40\ Gy}$ are the percent volume receiving dose ≥ 5, 10, 20, 30, and 40 Gy, respectively. [f]There is no reported IRR for the group ≥80% for $V_{40\ Gy}$ because of zero cases occurring in this bin. Hence, data were combined for 40% to <80% and ≥80%, which is represented by the IRR for ≥40% for $V_{40\ Gy}$. events, survivors with a colorectal SMN; IRR, incidence rate ratio; nonevents, survivors without a colorectal SMN; ref, reference; RT, radiation therapy; SMN, subsequent malignant neoplasm.

populations.[17,40] For dose-response relationships by chemotherapeutic groups, we observed that colorectal SMN rates increased with increasing cumulative doses of anthracyclines, alkylating agents, or platinum-based agents. For procarbazine, rates also increased with increasing cumulative dose in survivors with or without RT. Two previous studies reported a dose response with procarbazine. One study of colorectal SMN risk in survivors of primarily adult Hodgkin lymphoma[17] reported a rate ratio of 2.9 for procarbazine doses >8,400 mg/m². Another study of GI[6] SMN risk in childhood cancer survivors reported a hazard ratio of 3.15 for procarbazine doses ≥7,036 mg/m². Our study, focused only on colorectal SMN in childhood cancer survivors, also reported a dose response with procarbazine, but reported a higher risk (IRR, 9.0; 95% CI, 4.3 to 18.9 for ≥7,036 mg/m²) than either of those studies at the same/similar dose level. Of note, procarbazine dose ≥7,036 mg/m² had a rate of similar magnitude to a MCD ≥20 Gy (IRR, 8.3 [95% CI, 3.9 to 17.8]). We also reported that colorectal SMN rates increased linearly with procarbazine dose (ERR of 73.0% per 1,000 mg/m²). In unrestricted and restricted analyses, for nonprocarbazine alkylating agents, the yes/no effect was marginally significant, while the high-dose category (≥6,000 mg/m²) was significant. Of these agents, the most frequently used was cyclophosphamide followed by nitrogen mustard (mechlorethamine), which could be the focus of future studies (Data Supplement, Table S2).

Childhood cancer survivors who received ≥20 Gy MCD had >12-fold higher risk (SIR, 12.6 [95% CI, 6.7 to 21.5]) than the age-sex–matched general population and their AER was >1 additional case per 1,000 survivors annually. This group's absolute risk is approximately equal to the rate of the US population at age 60s for whom screening is already recommended. We observed increased rates for MCD 10 to <20 Gy (IRR, 3.6 [95% CI, 1.9 to 6.9]) and ≥20 Gy (IRR, 8.3 [95% CI, 3.9 to 17.8]), but not for 0.1 to <10 Gy (IRR, 1.1 [95% CI, 0.6 to 2.0]). Furthermore, we observed a dose-response relationship even in the absence of procarbazine (Fig 3B). While a recent case-control study,[17] among survivors of primarily adult-onset Hodgkin lymphoma, reported a linear dose-response between MCD and colorectal SMNs, to our knowledge, our study is the first to report an association in a pediatric cohort. Previous studies assessing the RT dose-response effect on colorectal SMNs in childhood cancer survivors were limited to prescription/delivered dose or local RT dose at the SMN site.[7,14] Here, we observed greater stratification of IRRs when assessed by MCD as opposed to abdomen and/or pelvic body region maximum target RT dose, that is, local prescription/delivered dose (Fig 3). For abdomen and/or pelvic body region RT doses, rates were elevated increased for doses of 10 to <20 Gy (IRR, 3.3 [95% CI, 1.5 to 7.2]) and ≥20 Gy (IRR, 3.3 [95% CI, 2.0 to 5.4]), but the difference between these dose groups was not statistically significant. By contrast, when IRRs were evaluated by MCD, the IRR for 10 to <20 Gy (IRR, 3.6 [95% CI, 1.9 to 6.9]) was 2.3-fold lower than for the dose group ≥20 Gy (IRR, 8.3 [95% CI, 3.9 to 17.8]). This observation highlights that colorectum-specific dosimetry is more individualized, accounting for the shape of the colorectum and regions of the colorectum shielded by patient-specific treatment field blocks.

To our knowledge, this is the first study to quantify the relationship between colorectal SMN risk and RT dose-volume metrics applicable in contemporary pediatric radiation oncology. Specifically, as the percent volume receiving ≥5, 20, 30, and 40 Gy increased, rates increased when the irradiated colorectal volume exceeded 20%. Rates markedly increased when large volumes of the colorectum received high doses, with IRRs as high as 11.8 for $V_{40\ Gy}$ ≥ 40% and 21.3 for $V_{30\ Gy}$ ≥ 80%. We also observed increased rates for high doses to small volumes (IRR, 5.3 for $V_{40\ Gy}$ of 0.1% to <20%). Two previous studies[8,13] assessed volume effects on colorectal carcinoma[8] or adenoma[13] risk using a surrogate exposure measure: the number of irradiated colorectum segments[8] or an assigned value of percent colorectum irradiated on the basis of RT field type.[13] Although these studies assessed volume effects, dose-volume metrics, which are commonly used in radiation oncology, were not explored.

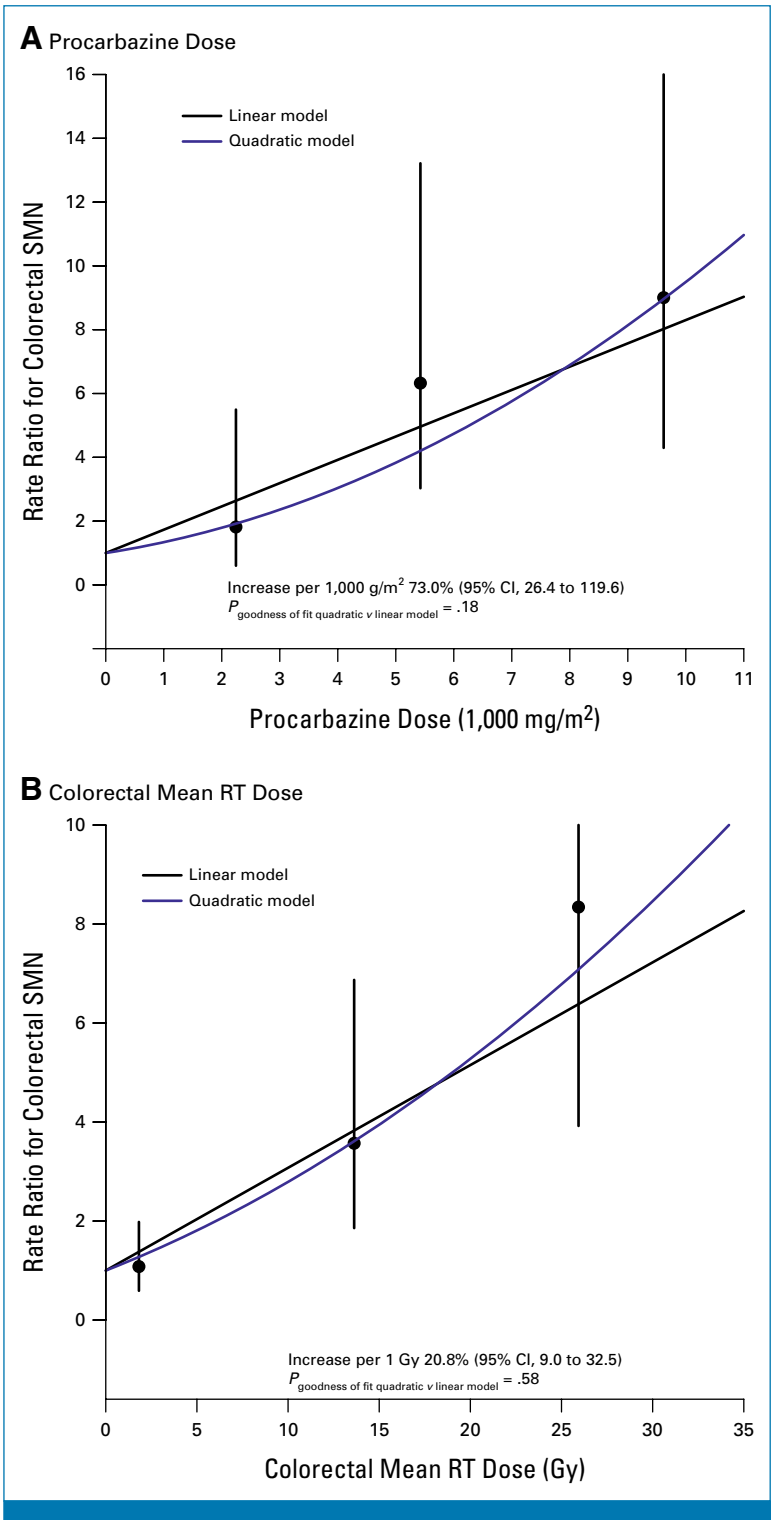


FIG 4. ERR dose-response models. ERR dose-response relationships between colorectal subsequent malignant neoplasm rates and (A) procarbazine dose and (B) colorectal mean RT dose. The procarbazine dose categories were <4,200, 4,200 to <7,036, and ≥7,036 mg/m^2 and the median procarbazine dose in these categories were 2,200, 5,400, and 9,600 mg/m^2, respectively. The colorectal mean RT dose categories were 0.1 to <10, 10 to <20, and ≥20 Gy, and the median colorectal mean RT dose in these categories were 1.8, 13.6, and 25.9 Gy, respectively. ERR models were adjusted for attained age, age at primary cancer diagnosis, sex, and race/ethnicity on the basis of a priori hypotheses. On the basis of univariate significance (P < .1), the procarbazine ERR model was also (continued on following page)

FIG 4. (Continued). adjusted for mean colorectal RT dose (cutoffs: none, 0.1 to <10, 10 to <20, and ≥20 Gy) and the colorectal mean RT dose ERR model was also adjusted for alkylating agents using cyclophosphamide-equivalent dose (cutoffs: 0, <6,000, and ≥6,000 mg/m^2). The filled circles and error bars represent the adjusted ERRs and 95% CI for the aforementioned dose categories and are plotted at each dose category's median value. We applied ERR models with a linear (black line) and quadratic (violet line) form. We compared the fit of the linear model with that of the quadratic model using a likelihood ratio test. There was no significant departure from linearity observed for procarbazine (P = .18) or for colorectal mean RT dose (P = .58). There was no statistically significant additive interaction between colorectal mean RT dose and procarbazine when considered as either yes/no (P = .26) or categorical dose (P = .72) variables. ERR, excess rate ratio; Gy, Gray; RT, radiation therapy; SMN, subsequent malignant neoplasm.

We did not observe a multiplicative interaction between RT and chemotherapy on colorectal SMN rates, which was also not seen in previous studies of survivors of primarily adult-onset survivors of Hodgkin lymphoma[17,31] and adult survivors of testicular cancer.[40] However, it is possible that our study, with 104 cases, was underpowered to detect such an interaction. Although an interaction was not observed, we observed that exposure to any alkylator, procarbazine, or platinum-based agent was associated with increased colorectal SMN rates in the absence of exposure to any RT (Fig 2B).

We observed that excess colorectal SMN rate increased linearly with MCD (ERR per 1 Gy 20.8%; 95% CI, 9.0% to 32.5%). Although our ERR estimate is higher than the 9.2% overall ERR per Gy (95% CI, 2.5% to 23.0%) reported for survivors of primarily adult-onset Hodgkin lymphoma,[17] the overlapping confidence intervals indicate that true risk may be similar. Solid cancer studies in other populations, for example, therapeutic radiation exposure[41] and at atomic bomb (neutron) exposure,[42,43] also indicate higher values for children exposed to ionizing radiation at younger versus older ages. For MCD, the quadratic ERR model did not improve the data fit compared with linear ERR models. This finding was also observed in an atomic bomb study.[43] A linear model suggests that there may be no threshold dose below which there is no added colorectal cancer risk from RT. Although the $V_{5\ Gy}$ piecewise-exponential model shows no risk below 20% volume irradiated (IRR, 1.0 for $V_{5\ Gy}$ of 0.1% to <20%), risk is elevated when moderate volumes 20%-40% (IRR, 2.8 [95% CI, 1.3 to 5.9]) and large volumes 40% to <80% (IRR, 3.9 [95% CI, 1.9 to 8.0]) and ≥80% (IRR, 5.2 [95% CI, 2.7 to 10.2]) receive ≥5 Gy. Taken together, these findings highlight the importance of reducing MCD and $V_{5\ Gy}$ in contemporary RT treatment planning as much as possible without compromising dose coverage of the tumor volume. Moreover, since contemporary RT techniques, such as intensity-modulated RT, increase low-dose radiation to surrounding normal tissue,[44] understanding colorectal SMN risk in low-dose regions (as quantified by the $V_{5\ Gy}$ model) may become more relevant for the growing population of survivors treated with contemporary RT techniques for whom empiric evidence on SMN risk is virtually lacking to date.[45-48]

As clinicians evaluate colorectal SMN risk in long-term survivors of pediatric cancer, the dose-response relationships by chemotherapeutic groups reported here are valuable new knowledge. However, colorectal-specific RT doses are not yet relevant in survivorship care because this level of dosimetry was not reported for survivors treated in earlier eras of RT. As such, colorectal SMN risk could be assessed on the basis of the most detailed RT data available in patients' medical records, which may include any RT, abdomen and/or pelvic RT, or total prescribed dose to the abdomen and/or pelvis. To meet that continued need, this study also reports IRRs on the basis of those metrics. Currently, on the basis of recommendations from the US Children's Oncology Group (COG), the only treatment exposures that result in early colorectal cancer screening are any exposure to abdominal, pelvic, spinal (lumbar, sacral, or whole), or total-body RT.[49] On the basis of our findings, it may also be important for exposure to any alkylating agent (IRR, 2.7 [95% CI, 1.6 to 4.4]) or any platinum-based agent (IRR, 3.2 [95% CI, 1.6 to 6.1]) to be considered in guidelines as we observed similar rates in multivariable analysis as was observed with any exposure to any abdominal and/or pelvic RT (IRR, 3.3 [95% CI, 2.1 to 5.3]). Additionally, our findings suggest that detailed dose-response data should also be considered to start early screening, such as high RT dose to large colorectal volumes ($V_{30\ Gy}$ > 80%), MCDs ≥10 Gy, alkylating dose ≥6,000 mg/m^2, and/or platinum-based dose ≥450 mg/m^2.

As for the age to initiate screening, the same rate (30.5 per 100,000) in which colorectal cancer screening is recommended for the general population at age 45 years[39] was observed at younger ages for the overall CCSS cohort (37.9 years), and for survivors who received any chemotherapy (37.6 years), alkylating agent (33.1 years), abdominal/pelvic RT (29.4 years), or procarbazine (25.6 years). Our findings for the RT risk group are in agreement with the current US COG guidelines,[49] which recommends screening 5 years after RT or at age 30 years (whichever occurs last). For

exposure to procarbazine, our results indicate that earlier screening (age 25.6 years) could be considered for survivors who received any procarbazine; however, studies with larger sample size are needed to further confirm the initiation of earlier screening on the basis of procarbazine exposure.

In contemporary radiation oncology, multiple different treatment modalities (eg, proton and photon) as well as image guidance and even motion management strategies are available and can be selected on a case-by-case basis to optimize RT with the aim of delivering the full prescription dose to the target volume while minimizing dose to organs at risk. Key to optimization is the definition of organ-at-risk dose constraints to minimize specific acute and late effects. In alignment with this common practice, associations between colorectal SMN risk and colorectal-specific RT dose and dose-volume metrics can be immediately translated into pediatric RT treatment planning. Beyond the keeping of colorectal RT doses as low as possible, constraints could be set to spare colorectum dose, for example, mean dose <10 Gy, $V_{20\ Gy}$ <20%, and maximum dose <40 Gy. When achieved, colorectal SMN risk may be reduced. When not achieved, these dose-volume metrics can be reported in the medical record and allow for future risk assessment. Moreover, a 2024 report from the Pediatric Normal Tissue Effects in the Clinic called for the need for dose-volume–based risk assessment, but also cited a lack of epidemiologic studies that included risk models on the basis of detailed organ-specific dosimetry.[46,50] This work is a first step toward filling that gap in knowledge.

In summary, this study provides a comprehensive report of colorectal SMN risk in childhood cancer survivors associated with cumulative doses of chemotherapeutic agents and the distribution of RT dose across the colorectum, that is, dose-volume metrics. Our findings are directly applicable to contemporary pediatric oncology practice and survivorship care, and of potential relevance for consideration in colorectal SMN surveillance guidelines as detailed RT dose-volume metrics become more available in the medical records of future survivors of childhood cancer.

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DATA SHARING STATEMENT

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of the CCSS data that leverage the expertise of CCSS Statistical and Survivorship research and resources will be considered on a case-by case basis. For this utilization, a research Application of Intent followed by an Analysis Concept Proposal must be submitted for evaluation by the CCSS Publications Committee. Users interested in utilizing this resource are encouraged to visit http://ccss.stjude.org. Full analytical data sets associated with CCSS publications since January of 2023 are also available on the St Jude Survivorship Portal at https://viz.stjude.cloud/community/cancer-survivorship-community~4/publications.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Colorectal-Specific Radiation Dose and Chemotherapy Risk for Subsequent Colorectal Malignancies in Childhood Cancer Survivors: A Childhood Cancer Survivor Study (CCSS) Report

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